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A System for Predicting Hospital Admission at Emergency Department Based on Electronic Health Record Using Convolution Neural Network

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***Abstract*— Emergency Department (ED) crowding has become an issue of delayed patient treatment and even a public healthcare problem around the world. According to recent research studies of many countries, the increasing number of patients in the emergency department which has led to unprecedented crowding and delays in care. For that reason, triage into five-level Emergency Severity Index (ESI) has become a major method for improving medical priorities in ED. Although the ESI mitigates the process of ED treatment, so far it still heavily relies on the nurse's subjective judgment and is easy to triage most patients to ESI level 3 in current practice. Therefore, a system that can help the doctors to accurately triage a patient's condition is imperative. In this work, we propose a system based on the patients' ED electronic health record to predict hospitalizations after assigned procedures in ED are completed. While most of the related studies have employed traditional machine learning for triage-related classification and highly relied on a feature selection process, our proposed system used data-to-image transform to produce the input and a convolutional neural network as a classifier. For validation, the data from an open dataset (National Hospital Ambulatory Medical Care Survey) is used which includes 118,602 patient visits of United States EDs from 2012 to 2016 survey years. To sum up, the resulting AUROC and accuracy achieve 0.86 and 0.77, respectively, in our work.**

Keywords—Emergency department triage; hospital admission; convolutional neural network

I. INTRODUCTION

Overcrowding in the emergency department (ED) is already a global public health issue and clearly an important patient safety issue[1] as well. Many countries, including Ireland, the United States, Canada, Germany, and Australia, have shown a continuous, significant increase in the number of EDs [2-7]. In the United States, ED visits were estimated to increase from 136.9 million in 2015 to 145.6 million in 2016, an increase of 6.4%. The 10-year volume change was 24.7% and has increased by a total of 61.2% over the past 20 years (the ED visits in 1996 were estimated at 90.3 million) [8, 9].

The increasing number of the ED visits has also caused periodic mismatches in the supply and demand of ED and hospital resources, which leads to longer waiting times and delays in critical medical treatments. The ED crowding is related with several adverse clinical outcomes, including higher mortality and morbidity [10, 11]. Therefore, the most important thing is to design a method to optimize the urgent patient's priorities in the ED [12].

The Emergency Severity Index(ESI) algorithm is a method used for ED triage, which can provide for classifying ED patients according to their acuity and resource requirements. The ESI can help doctors to decompose patients into five levels, with level 1 patients are the most urgent and must be cured first, whereas level 5 patients are the least urgent and can wait for a while. The goal of the ESI system is to differentiate more complex patients from those with simpler problems, thus improving patients throughput and treatment decisions [13, 14], including resource consumption rate, hospital admission rate, length of stay, and mortality. Over the past 20 years, many studies have been conducted to assess the validity, reliability, and ease of use of the ESI [12]. The ESI has also become the standard of a triage method for the ED, and it is currently the main implementation system in the United States [15].

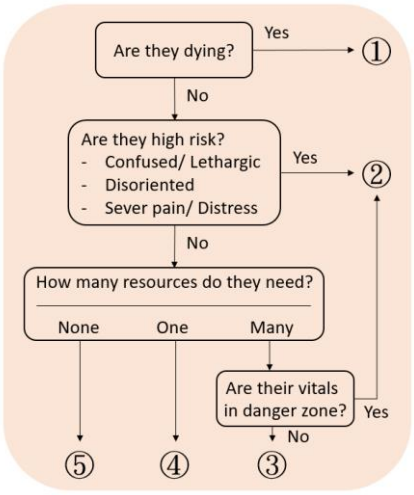


Figure 1: ESI Algorithm [16].

When the patients arrive at the emergency room, the nurse will check their vital signs, including blood pressure, temperature, heart rate, respiratory rate, *etc.*, and decide whether to further examine patients with X-ray, CT scan, MRI, Ultrasound and so on. During the process, the nurse will make a brief summary and record based on the pain content described by the patient. Finally, the triage level of patients is evaluated according to the ESI algorithm [16] in Figure 1.

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Despite the fact that the ESI is widespread adoption and ease of use, it still has several limitations, such as it relies heavily on the judgment and intuition of nurses [17, 18]. In [19], the authors got low accuracy of the ESI score when the nurses evaluated an ESI assignment for 30 standard written cases, although the nurses were high confidence in their ability of judgment. Moreover, the ESI cannot sufficiently distinguish patients by its five levels, most patients are triaged to ESI level 3 in United States, resulting in many undifferentiated patients, which undermine the purpose of triage [20]. This problem causes patients with multiple disease severity to gather in a large population, which may delay the care of critically ill patients. Moreover, the main purpose of ESI is to assign proper degree of urgency to patients and to allocate limited medical resources and hospitalization resources appropriately to patients in need.

TABLE I shows that the total 118,602 ED patient visits data collected from the National Hospital Ambulatory Medical Care Survey (NHAMCS) [9]. The table shows that the proportion of hospitalizations for patients with level 1 and 2 are much higher, while the proportion of hospitalization for patients with level 4 and 5 are relatively less. In ESI level 1 and level 2 patients, approximately 26% of them have a need for hospitalization, which also means that they have a very high demand for ED resources. In patients with level 3, there are relatively few people compared to patients with immediate urgent. Approximately 12% of the patients will eventually need to be hospitalized, but these people may have a chance to be triaged to Level 1 or 2 from the beginning. If those patients who have hospitalization need can be distinguished at early stage, they can avoid being triaged to less urgent level 3 and delayed treatment. In ESI Level 4 and 5, only approximately 2% of people are in need of hospitalization.

TABLE I. RATIO OF TRIAGE ESI LEVELS TO ADMISSION

	Triage ESI Level				
	Level 1	Level 2	Level 3	Level 4	Level 5
Admission	26.6%	26.4%	12.1%	2.5%	2.4%

Thus, this study will focus on establishing an effective and fast system for predicting whether patients will eventually be required admission in order to provide a reference for doctors to determine the priority ranking and treatment between patients in advance. In this proposed system, our goal is to use conventional structural data to design a binary classification model, which can distinguish the hospitalization needs of ED patient visits.

II. Related Work

Some research studies have focused on developing a system for predicting hospital admission based on the patient's ED electric electronic health record [21]. Among these studies, NHAMCS Dataset [22] is the most common dataset to be analyzed. Despite using the NHAMCS dataset, those works might have the different outcomes they achieved. Here, we will introduce some existing methods and implementation results, and briefly describe the concepts and methods that our system will use.

A. Existing approaches

In [23], Zhang *et al.* constructed a method for analyzing the patients' reason for visit by using principal component analysis and traditional natural language processing (NLP) combined with multilayer neural network models (MLNN) and logistic regression (LR). They tested the model by using ten-fold cross-validation, the area under the curve(AUC) of 0.84 was achieved in their work. In [24], Yan Sun *et al.* used the chi-square test to select the association between hospital admission and various possible risk factors, and input the extracted association features into logistic regression for training to develop a prediction model. They divided the dataset into derivatives (60%) and validation (40%), the AUC of 0.85 was achieved in their work. In [25], Graham *et al.* used three machine learning algorithms to create the models: logistic regression, gradient boosted machines (GBM), and decision trees. And the models were validated by using ten-fold cross-validation repeated five times, the accuracy of 0.8 was achieved. The purpose of their study can help plan the allocation of resources in advance and avoid the bottleneck of patient congestion. In [26], Hong *et al.* used deep neural networks (DNN), gradient boosting (XGBoost), and logistic regression to build different binary classifier models. The models were separately trained on three different information datasets, including triage information, patient historical information, and both of them. The information gain was then used to create low-dimensional indicators to judge variables of importance. They totally extracted 972 variables for each patient visit. The XGBoost model finally yielded the AUC of 0.92 in their work, and the result showed that the historical information improved model performance hugely compared to using triage information only.

B. Data to image

Regarding the data, there are many types in non-image form, including genomic, transcript, mutation, text, spoken words, financial, *etc.* In these fields, the prediction or classification of data is mainly using traditional machine learning algorithms. Assumption that the different kinds of data can be transformed from non-image form to a well-organized image form, more deep learning algorithms can be utilized, and it is possible to achieve higher classification performance. Therefore, there are many research studies [27-31] now trying to develop methods that can effectively reassemble and arrange non-image data, thereby applying the data in more fields.

C. Convolutional Neural Network

Convolutional Neural Network (CNN) [32-35] is a deep artificial neural network that is usually applied to analyze visual imagery. CNN has been proven to be very useful for object recognition and are well known as a powerful visual features extractor. The aim of CNN is to learn high-level features from the input data via convolutions. CNN consists of multiple layers of neurons and learnable hyper-parameters: weights and biases, which are defined as filters. CNN transforms the input data through convolutional and pooling layers with multiple filters at each stage producing multiple feature maps. After the flatten layer, the flattened output is fed to one or more fully connected layers for classification and

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backpropagation is applied to every iteration of training. Finally, the probabilities or scores of output class are generated.

III. METHODS

A. System overview

The system overview in Figure 2 shows that the system is separated into two parts, which are the training part and prediction part. The values of each patient's electronic health record are used as input, and the patient's hospital admission from the doctor are used as the ground truth. There are three steps in the part of model training: The first step is to pre-process the input data, such as selecting the features to be used and filtering the unusable or missing data. Next, the processed data will be transformed into the corresponding two-dimensional image, then the transformed image and ground truth will be used to train the binary classification model. Finally, the binary classification model is applied to predict the ED patient visits.

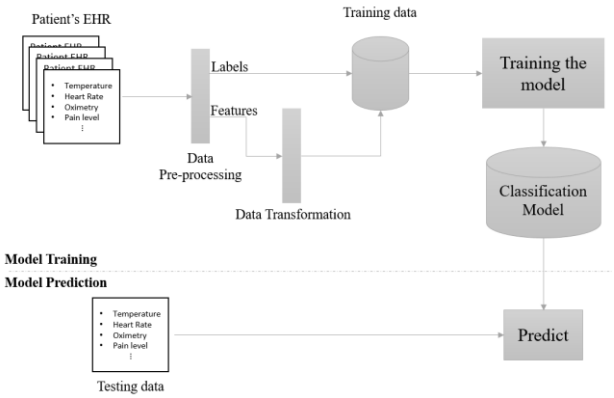


Figure 2. System Overview.

The proposed system can be utilized to classify a new visit for predicting either the subject will be hospitalized or not. The following sections will discuss the detailed methods of each part.

B. Data pre-processing

Since data are generated or transferred by human, data format errors or loss may occur. Therefore, it is necessary to pre-process the data through data cleansing to facilitate accurate data analysis. In order to ensure the accuracy and validity of the data, we need to first select the features to be used from the thousands of features in the dataset, and filter out the data with the wrong format, unusable or missing value.

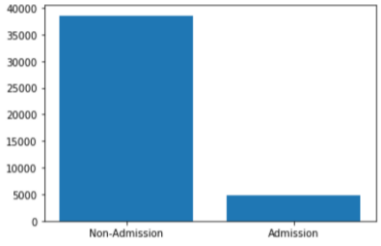


Figure 3. Data distribution.

Additionally, there is an imbalance problem [36] in the selected data. The number of patients who are not hospitalized is eight times more than the number of patients who are hospitalized, as shown in Figure 3. In this case, the model tends to be biased towards the patients who are not hospitalized during training, and will show a very poor classification rate for the patients who are not hospitalized. Therefore, we decided to random under-sample the data first, and then set the class weighted according to the amount of different data.

C. Proposed CNN model

After pre-processing the data, we converted the selected data from one dimension to two dimensions with reference to [27], and resized the two-dimensional data to 120×120, then normalized the image and inputted it into the model.

In our classification model, as shown in Figure 4, we chose the convolutional neural network algorithm to train the model. CNN is a common deep learning technique, which can be formulated as:

$$y = \mathrm{CNN}\,(I) \tag{1}$$

where I represents the input selected data and y represents the output label. The architecture of CNN is mainly constructed by three different types of layers, including convolutional layer, pooling layer, and fully connected layer. Each layer of CNN can transform a three-dimensional volume of activations to another three-dimensional volume of activations, and obtain the spatial and temporal dependence of the image through relevant filters. Then input the learned features into the fully connected neural network to train the classification model.

First, we trained the model with training data $\mathrm{I} \in \mathbb{R}^{H \times W \times C}$, where H is the height of the image, W is the width of the image, C is the channel of the image, $\mathbb{R}$ represent to encode the image into the latent space. The extracted function E condenses the input image into a low dimensional D by ignoring the noise and modeling the data distribution. The part can be formulated as:

$$D = E(wI + b) \tag{2}$$

where w, b are the weights and biases of the parameters of the convolutional layer. The model is composed of four convolution neural network layers with 4×1 kernel size and all layers are followed by a max polling with 2×2 kernel size.

$$f(x) = \max(0, x) \tag{3}$$

Each layer of convolution neural network uses rectified linear unit (ReLU) non-linear activation function (3), which enables better training of deeper networks. The sigmoid function (4) is used to limit the output to the range of 0 to 1 in the output layer, and the effect is better when the feature difference is complicated or is not very large. Here, the sigmoid function will turn an image vector x into a probability value p, which is often utilized to evaluate the probability of each class .

$$p(x) = S(x) = \frac{1}{1 + e^{-x}} \tag{4}$$

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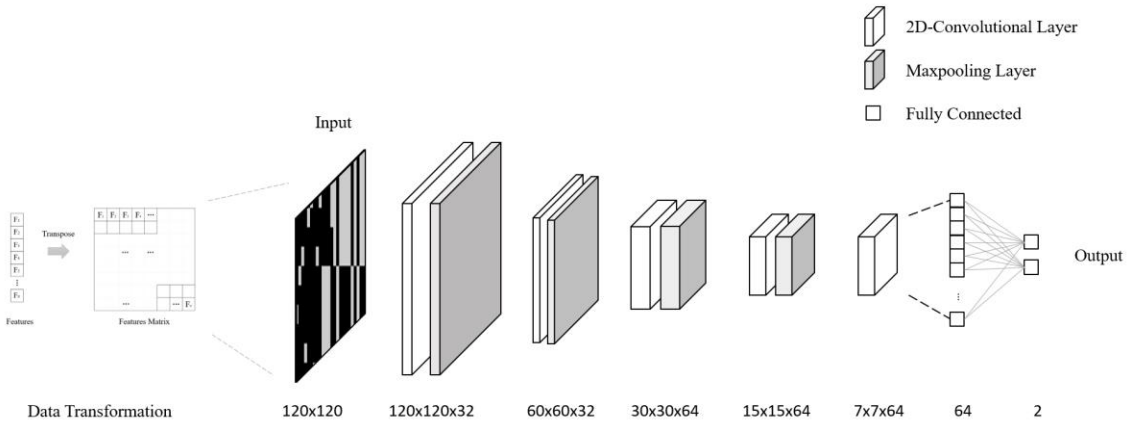


Figure 4. The architecture of our model.

The cross-entropy (5) is utilized to evaluate the difference in probability distributions. Here, the cross-entropy loss function L is utilized to measures the performance of a classification model to gain a loss value by comparing the ground truth y and the output prediction p.

$$L(x) = -\sum y(x)\log p(x) \tag{5}$$

The optimizer computes individual learning rates for different parameters and adjusts the trainable parameters or weights to minimize the loss of the batch.

IV. EXPERIMENT

In order to evaluate the effectiveness of the model, all experiments will be followed by below two steps:

- To completely mix the two groups of data, the data will be randomly shuffled.
- In each iteration, we split 80% of the dataset as training data and use it to train the model, and 20% of the dataset is used as the test data.

A. NHAMCS Dataset

The dataset is a review of the National Probability Sampling Survey conducted by National Center for Health Statistics during the patient visits to hospital outpatient and emergency departments and hospital-based ambulatory surgery centers. The NHAMCS has three main parts: hospital emergency departments, outpatient departments, and ambulatory surgery locations. The researchers visited more than 400 hospital-based EDs and collected the dataset within a random 4 weeks of each year. Data processing and medical coding were conducted by RTI International, Research Triangle Park, North Carolina. Although the statistics data and items provided by the NHAMCS dataset are slightly different in each year, the features we choose are provided in the statistical data every year.

In our study, 118,602 ED patient visits data during 2012 to 2016 were used. The selected feature items are in the triage stage, including residence, gender, age, arrival method and time, insurance, heart rate, blood pressure, temperature, respiratory, oximetry, triage level, pain level, *etc*. Then we transformed and normalized the selected data.

B. Experiment platform

We used Keras to execute all algorithms on computers with Nvidia GeForce GTX 1080 Ti GPU (with 11GB RAM) and Intel Core i5 CPU (with 32GB RAM). In the processing of the loss function, we used the Adam optimizer with learning rate of 0.0001 and a batch size of 32. The dropout rate was 0.5.

C. Evaluation criteria

In this study, we used two metrics to analyze the performance of our proposed model, including accuracy and area under the receiver operator characteristic curve (AUROC). Accuracy is used to show the correct rate of classified ED visits and determine whether they are actually classified into corresponding class. The receiver operating characteristic curve used to evaluated the relationship between true positive rate and false positive rate. Sensitivity (6) refers to the ability to detect ED visits who will be hospitalized finally. Specificity (7) relates to the ability to distinguish ED visits who will not be hospitalized. Precision (8) is used to assess how many of those who we predict as hospitalizations are actually hospitalized.

$$TPR(Sensitivity) = \frac{TP}{TP + FN} \tag{6}$$

$$TNR(Specificity) = \frac{TN}{TN + FP} \tag{7}$$

$$PPV(Precision) = \frac{TP}{TP + FP} \tag{8}$$

where $TP, TN, FP,$ and FN are true positive, true negative, false positive, and false negative.

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D. Baseline and result

Since most existing studies used different data sources, in order to effectively evaluate the prediction quality of our model, we chose traditional machine learning algorithms commonly used in other studies as our baselines, including logistic regression, XGBoost, and random forest. Then compare the differences in the results between different methods in various metrics. In this section, the performance on different models were evaluated. We compared our proposed CNN model with baseline machine learning algorithms that can learn from structural data.

TABLE II. PERFORMANCE ON DIFFERENT MODELS

Metrics / Methods	Recall	F1-score	Precision	AUROC
Logistic Regression	0.731	0.746	0.746	0.826
XGBoost	0.733	0.748	0.748	0.829
Random Forest	0.738	0.744	0.745	0.826
Our model	**0.790**	**0.790**	**0.820**	**0.860**

TABLE II shows the four measurements of each algorithm. It can be seen that our CNN model outperforms all the other algorithms. Our model can achieve the AUROC of 0.86 and the recall (sensitivity) of 0.79. The F1-score and the precision reached 0.79 and 0.82, which are also relatively higher comparing with the performance of other algorithms. The F1-score is based on the precision and recall of the test. The higher the F1-score, the better the performance of the model. Confirmed by the above methods, our deep learning algorithm improves results than traditional machine learning algorithms.

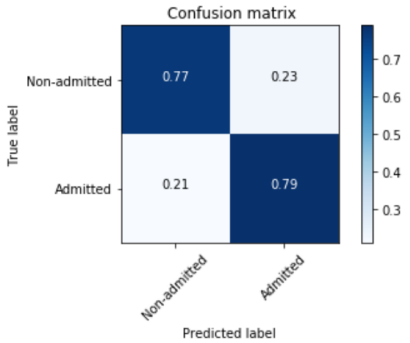


Figure 5. Confusion matrix.

The confusion matrix of our model is shown in Figure 5. The recall achieved 0.79 in our work, and 0.77 for specificity. According to two measurements above, our model is average and effective in the result of binary classification. According to the results of the final prediction of hospitalization, it is not biased to a specific class, and this strongly shows that our design can overcome the problem of uneven distribution of data.

E. Compare with other studies

According to existing research on the prediction of hospitalization demand, most of the current methods used specific feature selection methods combined with traditional machine learning algorithms, but due to the problem of uneven data, the final classification results may be controversial.

TABLE III. PERFORMANCE OF DIFFERENT RESEARCH STUDIES

Models	TPR	NPR	Acc	AUROC
NLP+PCA+LR[23]				0.846
Chi-square+LR[24]				0.849
GBM[25]	0.538	0.904	0.803	0.859
CNN (our model)	0.790	0.771	0.775	**0.860**

As shown in TABLE III, the performance is compared with other studies. Although the accuracy or AUC is high, assumption that the model is trained with unbalanced data, it may be affected by a specific class data. Therefore, in order to effectively train the model, the more balanced the data, the higher the model's TPR and NPR, and the better the performance. In addition, since the above research studies only [23] uses the same dataset, the others use private datasets. In order to fairly evaluate the performance of the model, according to the results of the previous comparison with traditional machine learning algorithms in TABLE II, it can be expected that if our method is applied to other research private datasets, it may be better results.

V. CONCLUSION

Emergency department (ED) crowding has become one of the biggest issues in medical services. Many countries have shown a continuous, significant increase in the number of ED's patient visits. Although the ESI system mitigates the process of treatment, it still relies on the nurse's judgment and is prone to triage most patients to ESI level 3. Moreover, the main purpose of ESI is to group patients and provide limited resources for those who need them. Therefore, a system that can help the doctors to accurately triage a patient's condition is imperative. In our work, we proposed a system based on the patients' ED electronic health record to predict hospitalization needs after assigned procedures in ED are accomplished. The system was based on a convolutional neural network and used data-to-image transform to produce the input to classify people into two groups, of which one needs hospitalization whereas the other does not. We have validated the model on an open dataset, called NHAMCS, containing 118,602 ED visits. As a result, the AUROC and the accuracy achieved 0.86 and 0.77. Although the dataset doesn't include the detailed text information of the patients when they arrived at the ED, the results of this study still can show the effectiveness of our model. For future work, we will use the local private dataset that we are collecting and apply unstructured data to this system to provide a more accurate and comprehensive triage performance.

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